

Interpretable and generative models for mouse spaceflight

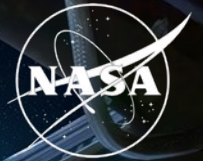
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Biological & Physical Sciences

National Aeronautics and
Space Administration



PROJECT

One dataset, two machine-learning questions

We used mouse bulk RNA-seq to study how spaceflight changes genes, pathways, and tissue state.

NASA OSDR API Mus musculus bulk RNA-seq | spaceflight (FLT) and ground control (GC) | multiple missions and tissues

01 Which biological programs change?

expiMap

Train a pathway-constrained reference on ARCHS4, then map OSDR samples into mouse Reactome programs.

Output: tissue-specific pathway shifts

02 Can synthetic profiles sharpen the analysis?

Conditional generation

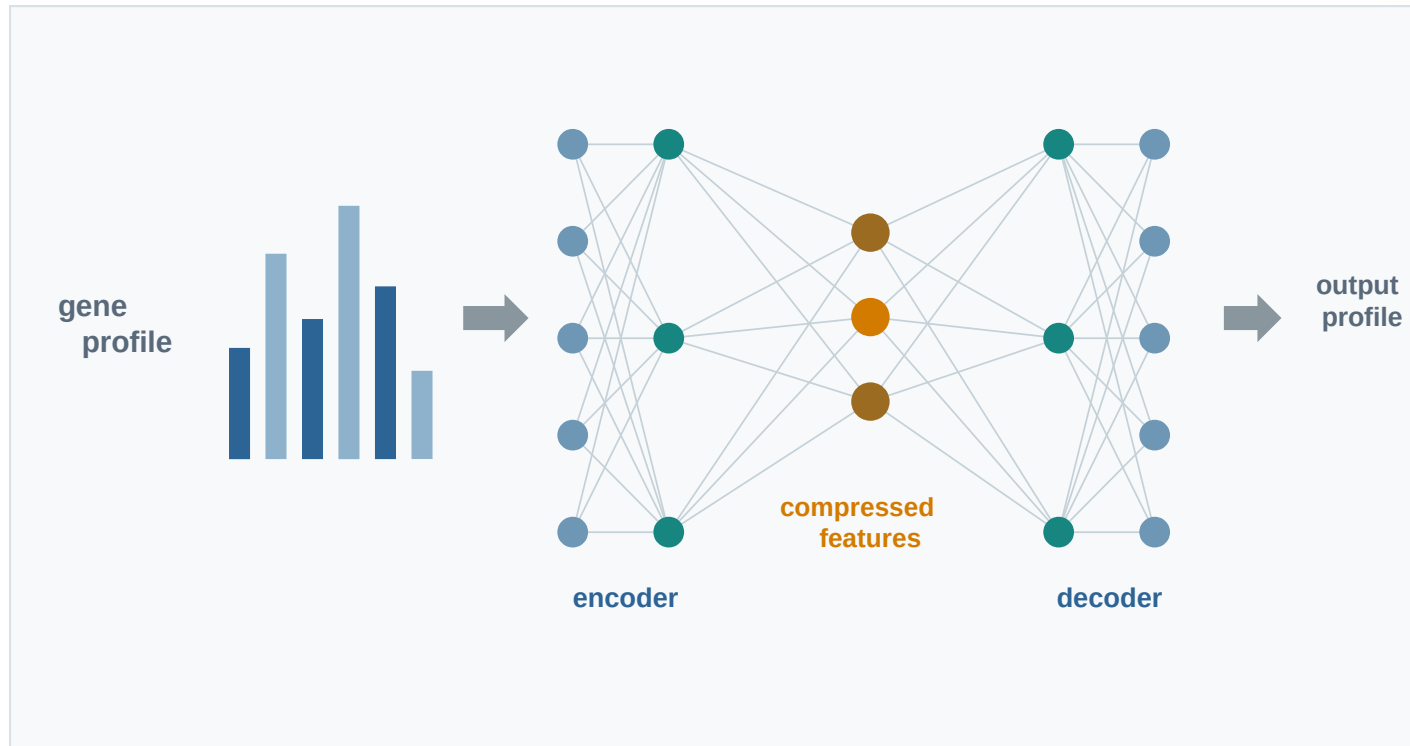
Generate condition-specific expression profiles by tissue, study, and FLT or GC condition.

Output: validated profiles and gene ranking

Autoencoders compress thousands of genes into a few features

The encoder summarizes a gene profile; the decoder learns enough structure to reconstruct it.

AUTOENCODER



COMPRESSED SPACE



expiMap: each compressed feature corresponds to a Reactome gene program.

EXPIMAP SCORES

Program scores turn expression into pathway-level changes

Each sample receives a Reactome program score; FLT minus GC shows whether the score rises or falls in flight.

REPRESENTATIVE OBSERVED OSDR PROGRAMS					
		FLT score	-	GC score	= FLT - GC
Thymus	DNA repair	-0.45	-	+0.55	-1.01
Skin	Cell-cell junction organization	-0.09	-	+0.11	-0.20
Liver	MHC class II antigen presentation	-0.07	-	+0.09	-0.16
Spleen	T-cell receptor signaling	-0.13	-	+0.13	-0.26

■ lower □ no shift ■ higher

HOW TO READ IT

- 1 Score each sample**
One score summarizes the genes in a Reactome program.
- 2 Compare within study**
Flight samples are compared with ground controls from the same study.
- 3 Combine study results**
The study-level changes are combined across missions.

Negative change = lower program score in flight

EXPIMAP RESULTS

Four tissues produced the clearest pathway patterns

FLT and GC were compared within each study before the study-level changes were combined.

Thymus

117 samples | 5 projects

LOWER IN FLIGHT

DNA repair; RHOA cytoskeletal cycle; lymphoid-stromal interactions

Known involution may also involve lower repair, motility, and niche coordination.

Skin

151 samples | 4 projects

LOWER IN FLIGHT

Chromatin regulation; DNA repair; Hedgehog; sphingolipids; cell junctions

Barrier injury may include a broader loss of tissue maintenance and coordination.

Liver

197 samples | 9 projects

LOWER IN FLIGHT

MHC class II antigen presentation; T-cell receptor signaling

Metabolic heterogeneity coexisted with lower adaptive immune communication.

Spleen

100 samples | 5 projects

LOWER IN FLIGHT

T-cell receptor signaling; neutrophil degranulation; C-type lectin signaling

The most consistent multi-pathway result joined adaptive and innate immune changes.

BACKGROUND

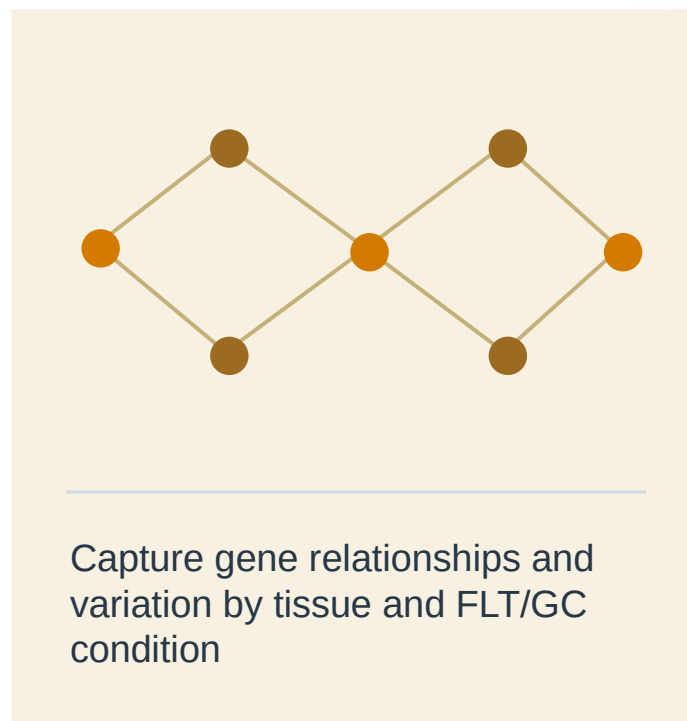
What is synthetic transcriptomics?

A generator learns patterns in measured RNA-seq data, then samples new expression vectors under chosen conditions.

01 Observed profiles



02 Learn expression patterns



03 Synthetic profiles



QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

Observed spaceflight cohort

1,610

profiles

75

OSDR studies

835 FLT

775 GC



ARCHS4 mouse

Reference pretraining cohort

997,515

profiles screened



17,244

selected

20 tissues

Study effects can look like spaceflight biology.

- 01 Preserve tissue and FLT/GC structure.
- 02 Avoid memorizing individual profiles.
- 03 Test biological effects in observed OSDR samples.

HOW THE GENERATORS WORK · 1 OF 2

Diffusion: denoise from noise

Start from pure noise and denoise step by step onto the data manifold, conditioned on FLT or GC.

FAMILIAR · IMAGE DIFFUSION



OURS · GENE-EXPRESSION DIFFUSION

reverse diffusion

noise

step $t = 1000$ 

● FLT ● GC

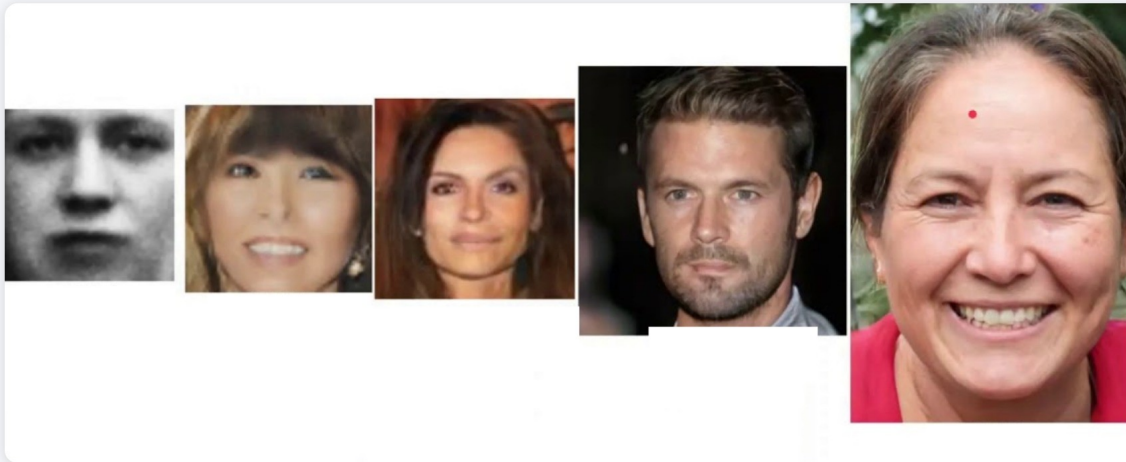
each condition lands on its own spiral

HOW THE GENERATORS WORK · 2 OF 2

Conditional WGAN-GP: generator versus critic

A generator and a Wasserstein critic compete until fake profiles are indistinguishable from real.

FAMILIAR · GANS FOR FACES



OURS · PROFILE GENERATOR

adversarial training: G vs critic



G trains until the critic
can no longer tell them apart

METHOD

We built a configurable bulk RNA-seq generation pipeline

We tested alternative processing, study correction, training, and conditioning choices.

01 Data scope

Sources

OSDR API

ARCHS4

Studies

Single

Multiple

Tissues

Per tissue

All tissues

02 Processing

Expression

Raw

CPM

TPM

Scaling

None

Z-score

Range
scale

Features

All

Variable

974
genes

03 Study effects

Alternative methods

None

Within-study z-score

ComBat / MBatch

MOBER

04 Model training

Generator

WGAN-GP

DDIM

Training source

OSDR only

ARCHS4 only

ARCHS4 then OSDR

05 Conditioning

Model inputs

Tissue

FLT / GC

Study

Material type

Sex / age

Downstream

configuration

PROCESS

TPM / scaled / 974 genes

PRETRAIN

ARCHS4

ADAPT

OSDR

GENERATE

Conditional DDIM

CONDITION

Tissue

FLT / GC

Study

Material type

HARMONIZE

None

SCOPE

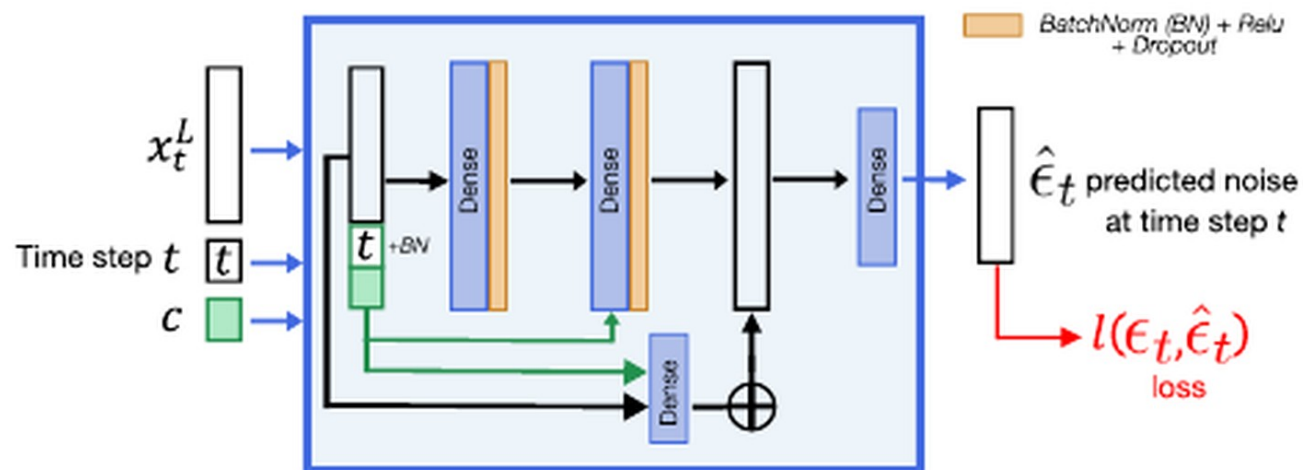
All tissues

VALIDATION

Diffusion best reproduced the measured expression distribution

Both models reproduced gene expression. Diffusion samples were harder to distinguish from measured samples.

Architecture from Lacan et al.



The network repeatedly predicts and removes noise using tissue and time information.

For OSDR, the model also receives FLT/GC, study, and sample-material information.

Tissue identity retained

Real 0.781 | Synthetic 0.781

Balanced accuracy is preserved in generated tissue labels.

Metric	WGAN-GP	DDIM	Read
Expression corr.	0.976	0.974	higher
Coverage (F1)	0.985	0.997	higher
Real-vs-synth. acc.	0.636	0.475	0.5 is ideal
Distribution distance	0.144	0.074	lower

Use diffusion

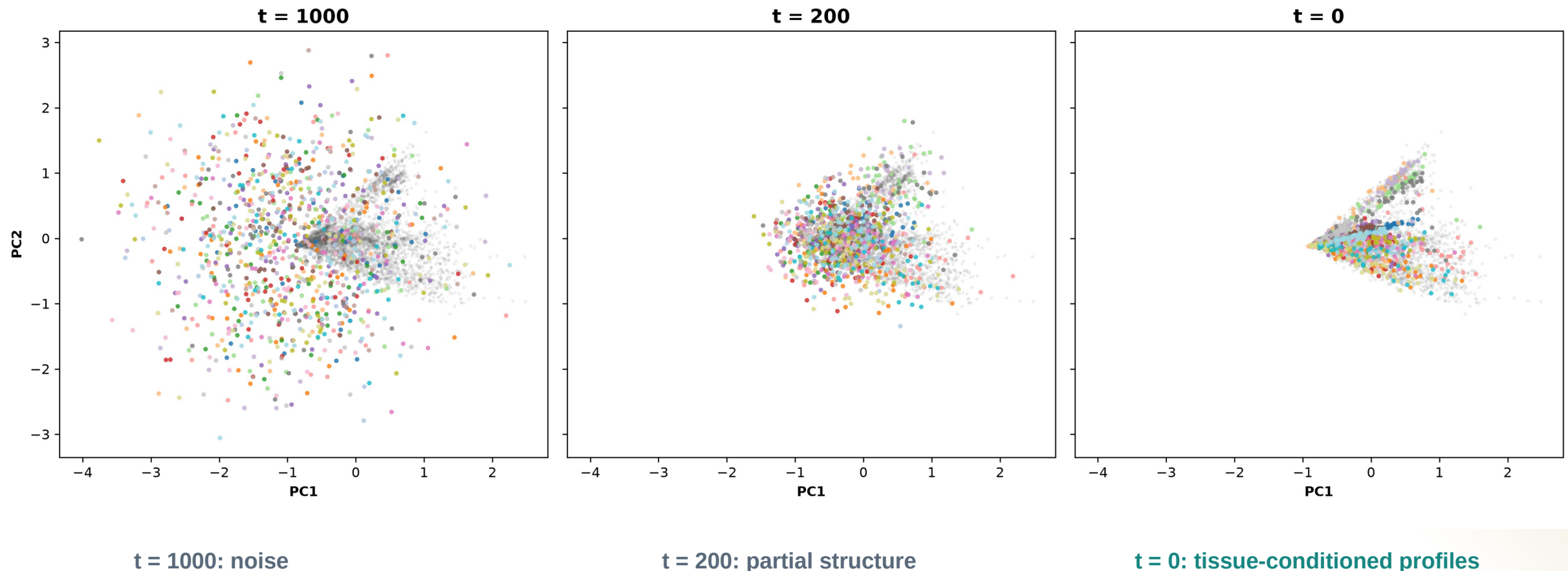
Better coverage and closer to the real distribution.

A score near 0.5 means a prediction model cannot reliably tell measured and generated samples apart.

DIFFUSION

Diffusion learns tissue structure from noise

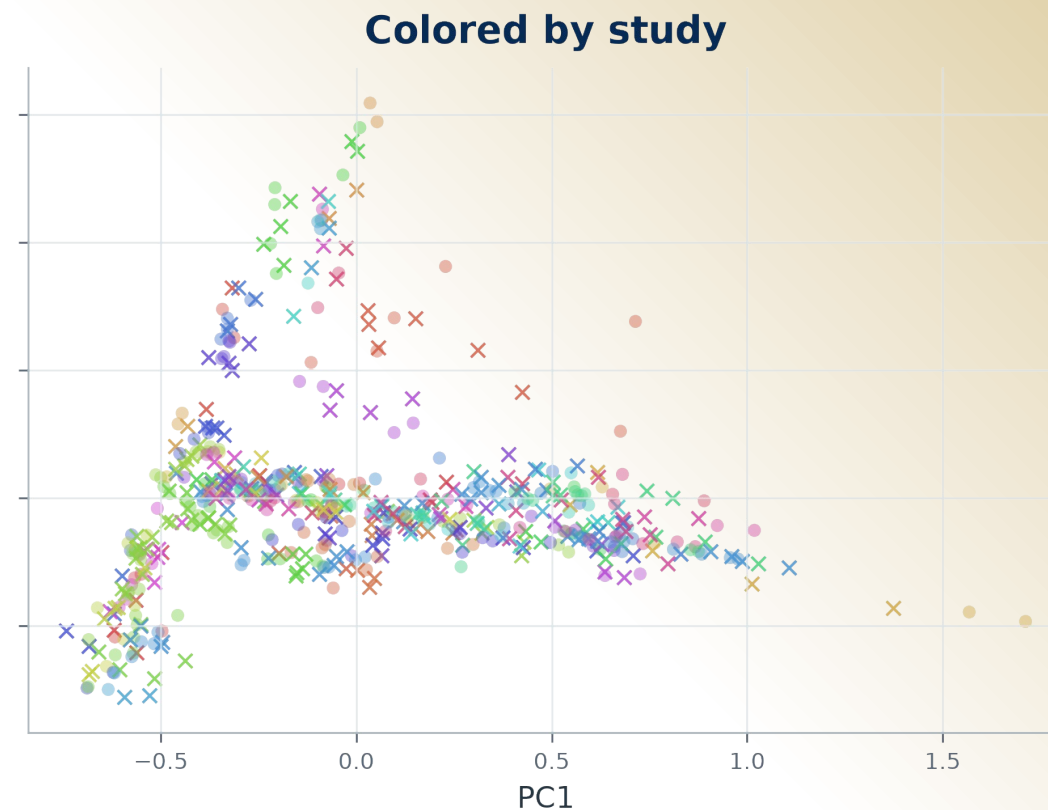
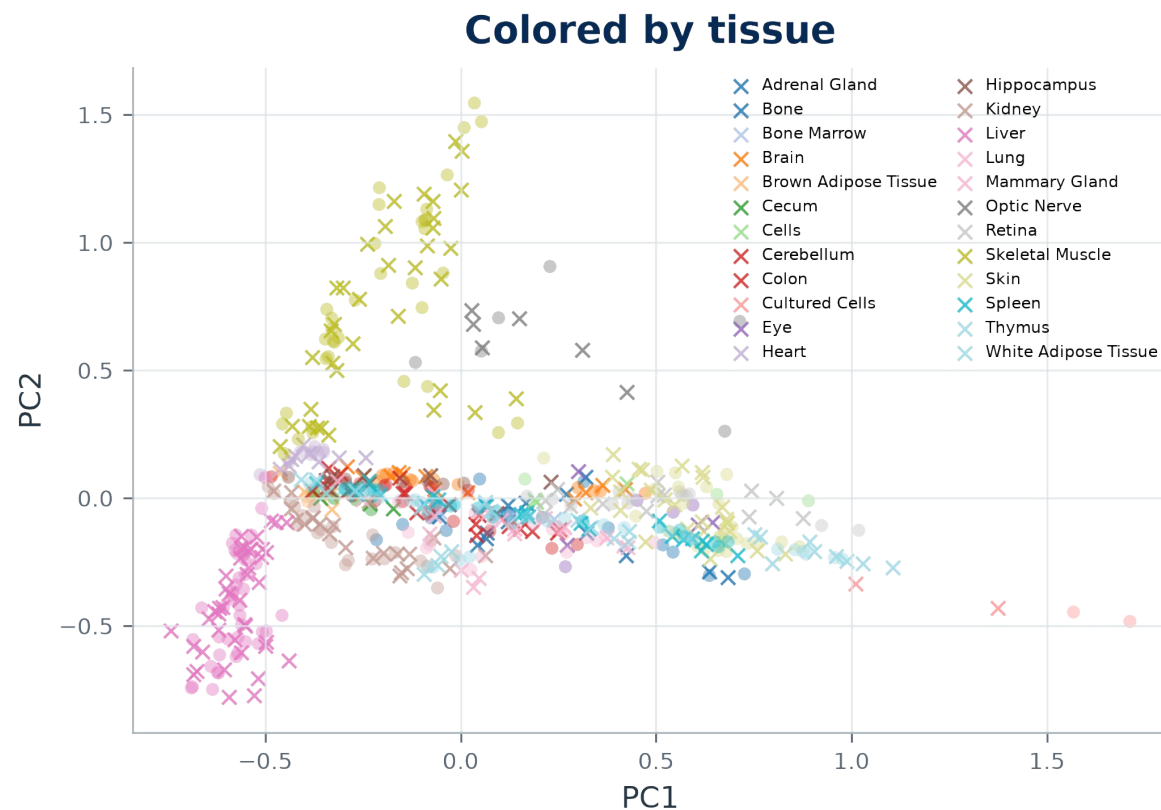
The same PCA axes follow 1,024 generated profiles through reverse diffusion.



DIFFUSION OUTPUT

Tissue and study structure dominate the PCA space

Left: tissue. Right: study. Circles are measured samples; crosses are generated samples.

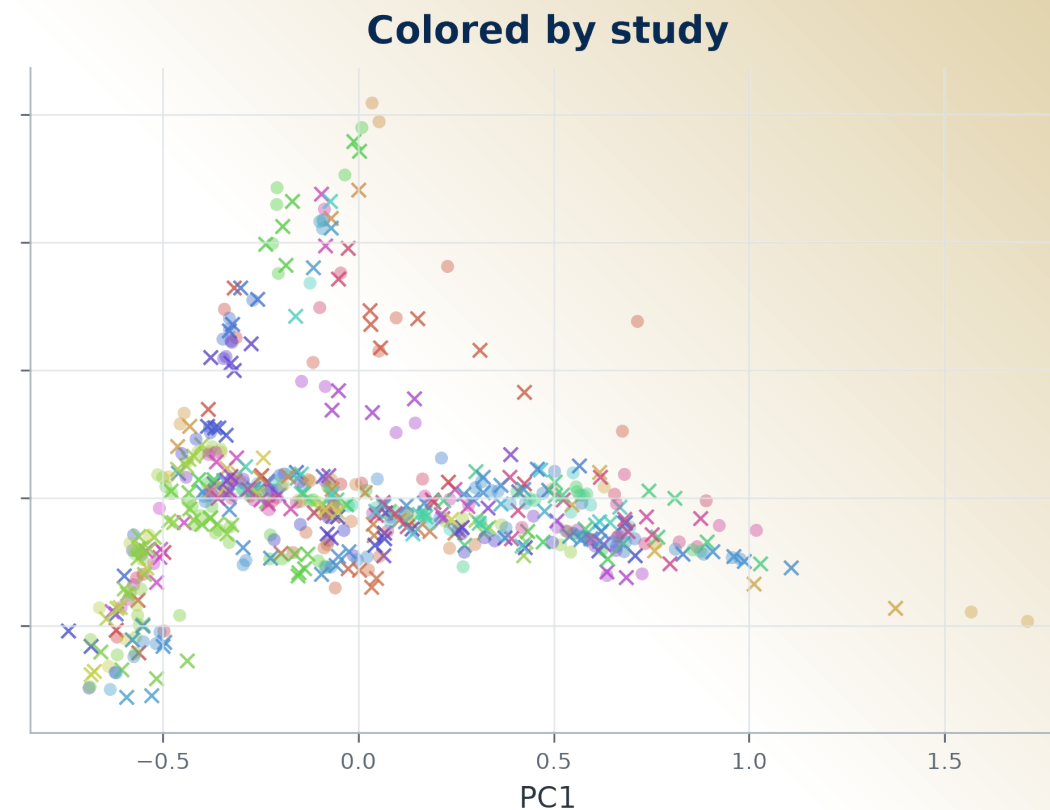
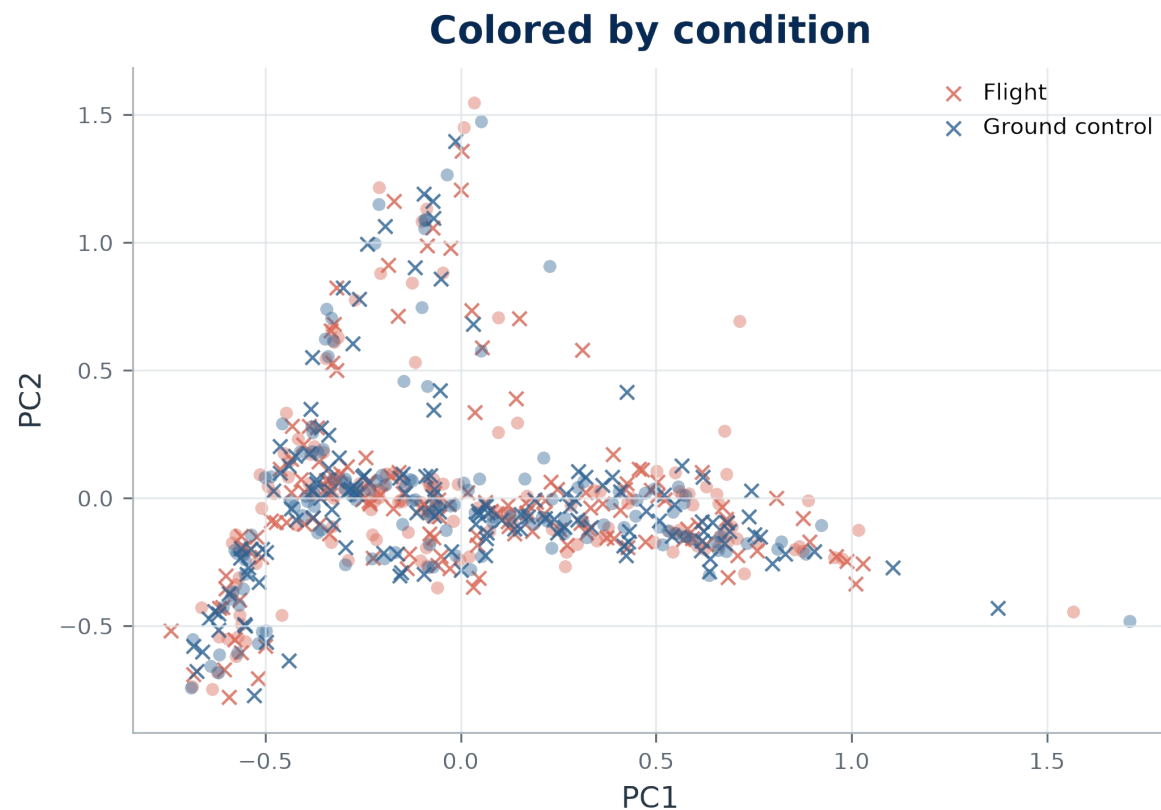


Generated samples reproduce the tissue and study patterns in the measured data.

DIFFUSION OUTPUT

Flight condition is subtler than study structure

Left: FLT or GC. Right: study. Circles are measured samples; crosses are generated samples.

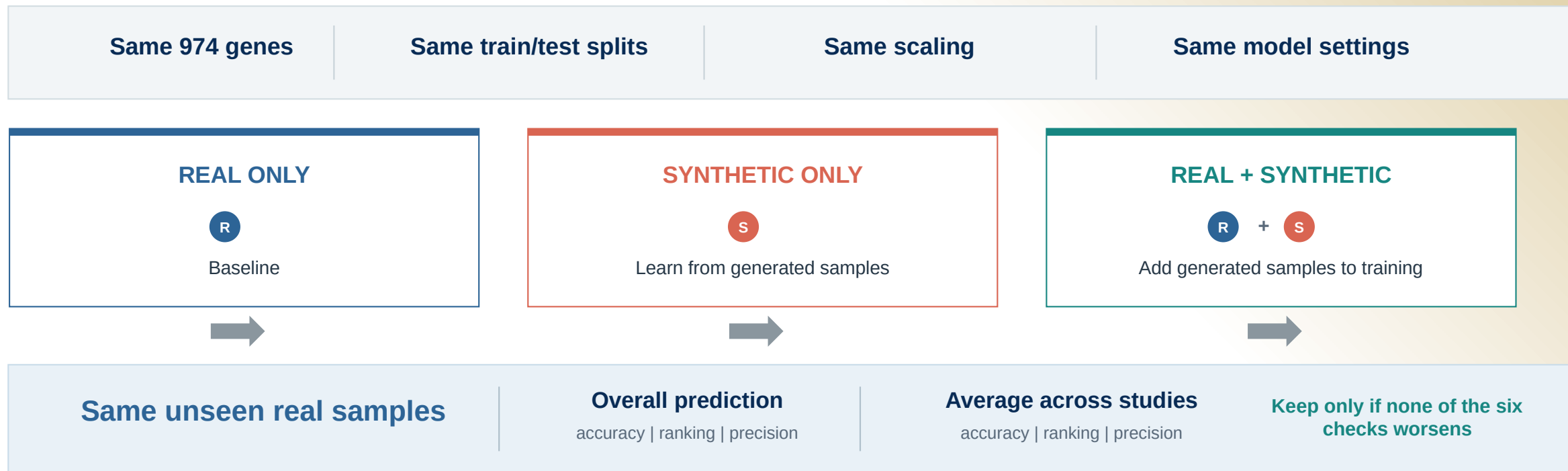


FLT and GC overlap more than samples from different studies do.

PRIMARY ANALYSIS

A fair test changes only the training data

Each prediction model uses the same genes, processing, train/test splits, and settings.

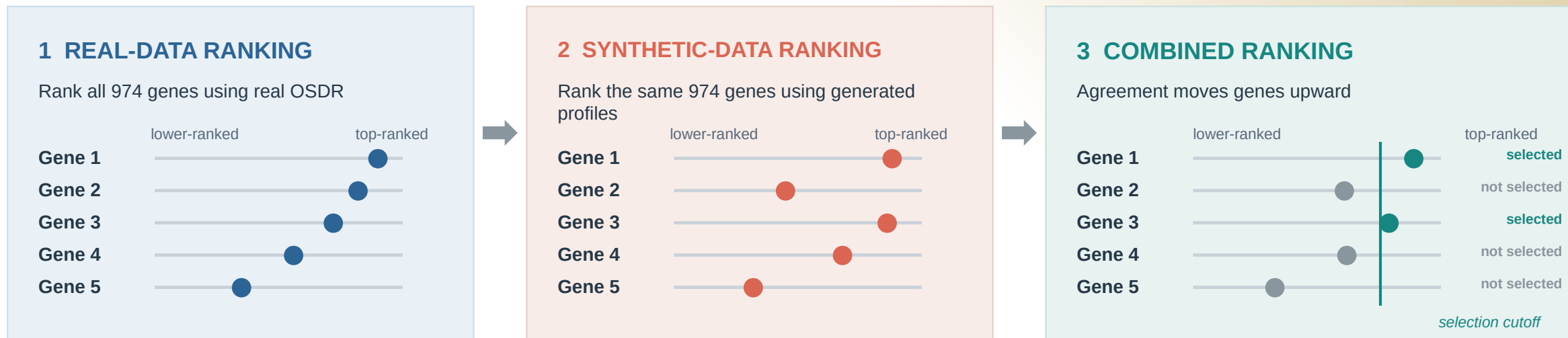


Which genes helped? Measure each gene's contribution on unseen real samples

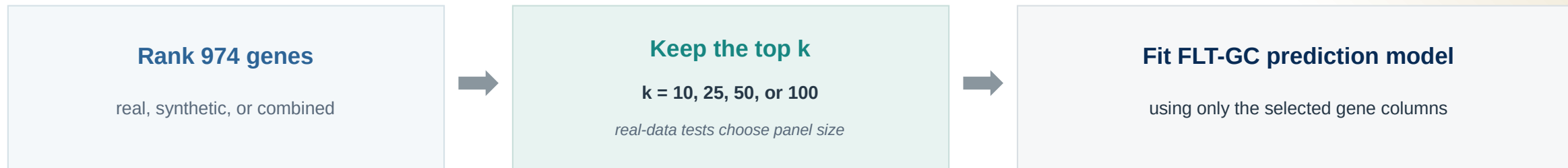
Significance comes from real OSD data

Combine real and synthetic rankings to choose a small gene panel

Genes supported by both datasets rise in the combined ranking; only the highest-ranked genes enter the FLT-GC model.



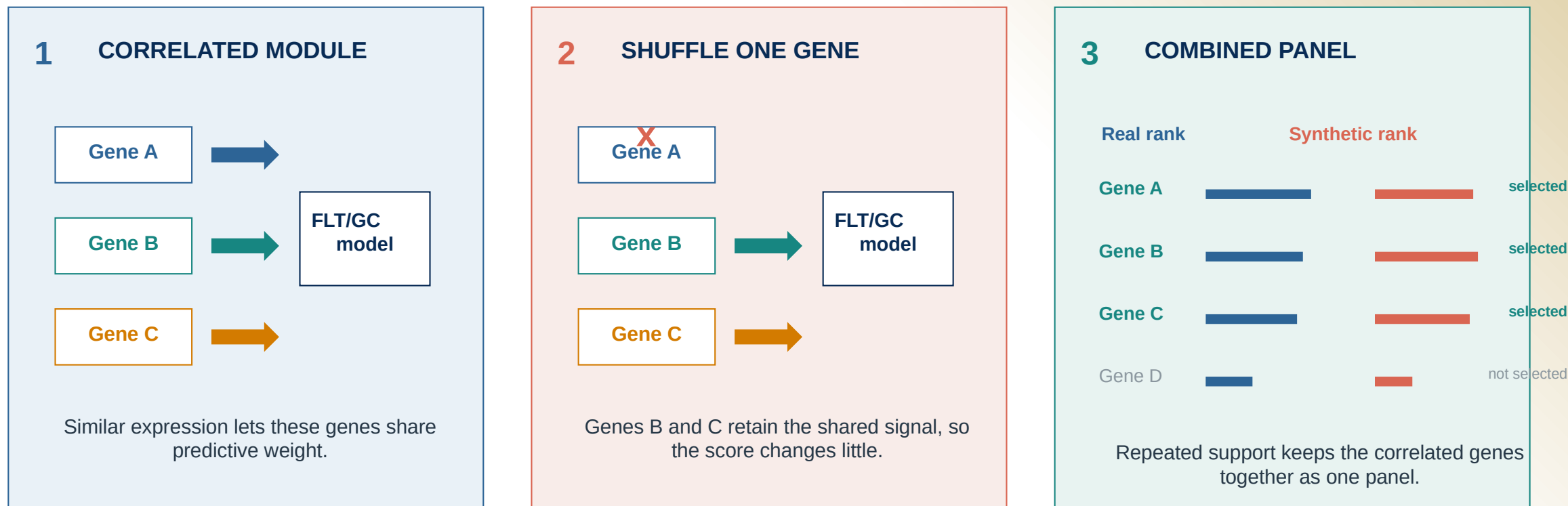
Top-ranked genes become model inputs



INTERPRETATION

Correlated genes can hide one another in one-at-a-time tests

A pathway can matter even when shuffling one gene causes only a small drop in prediction.



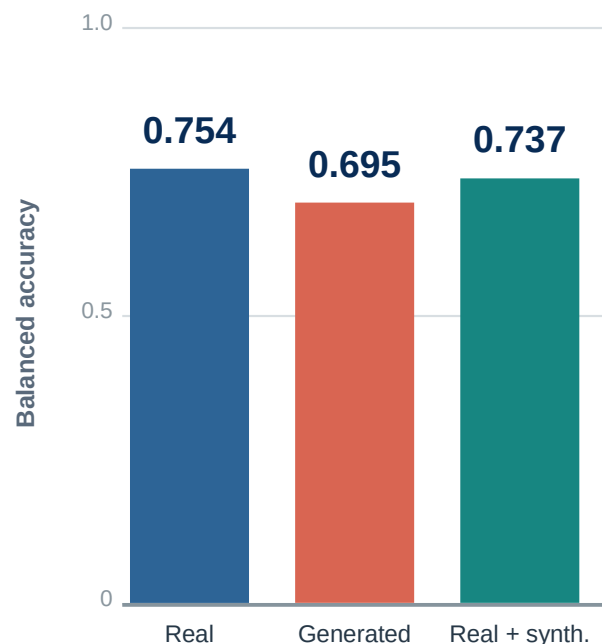
A low one-gene score can mean that another correlated gene carries the same information. It does not make the gene irrelevant.

PREDICTION

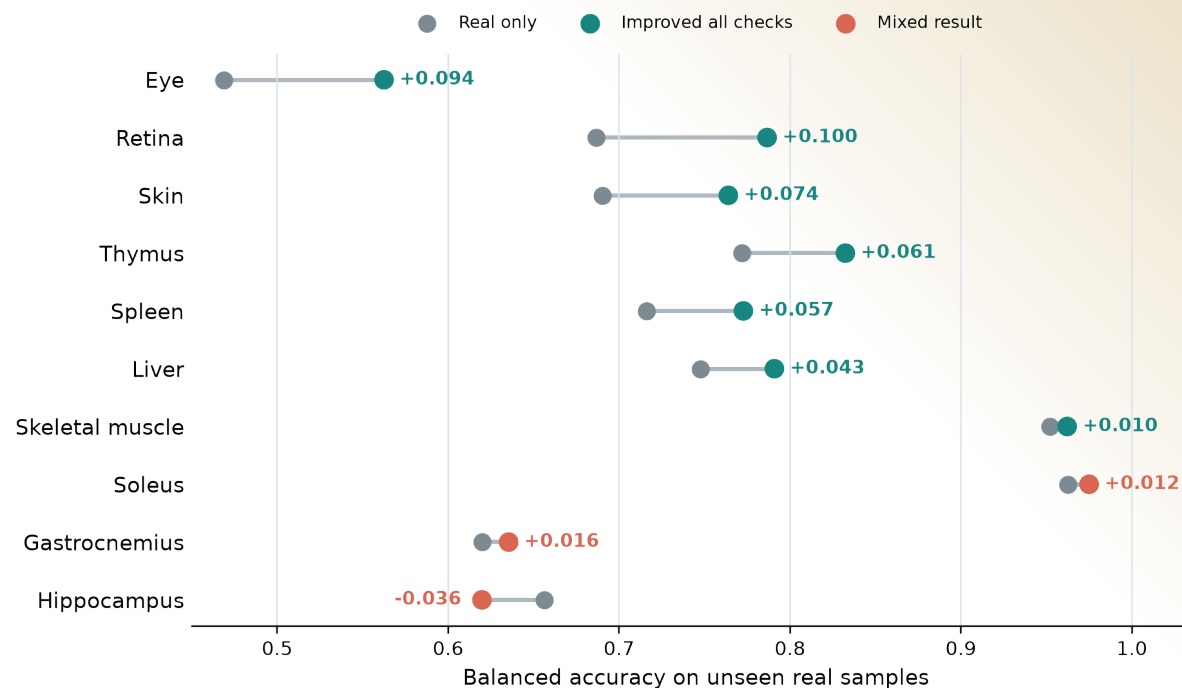
Adding synthetic samples helped many tissues, but not all

We kept the genes and model settings fixed. Only the training data changed.

One model across all tissues



Real + synthetic, analyzed by tissue

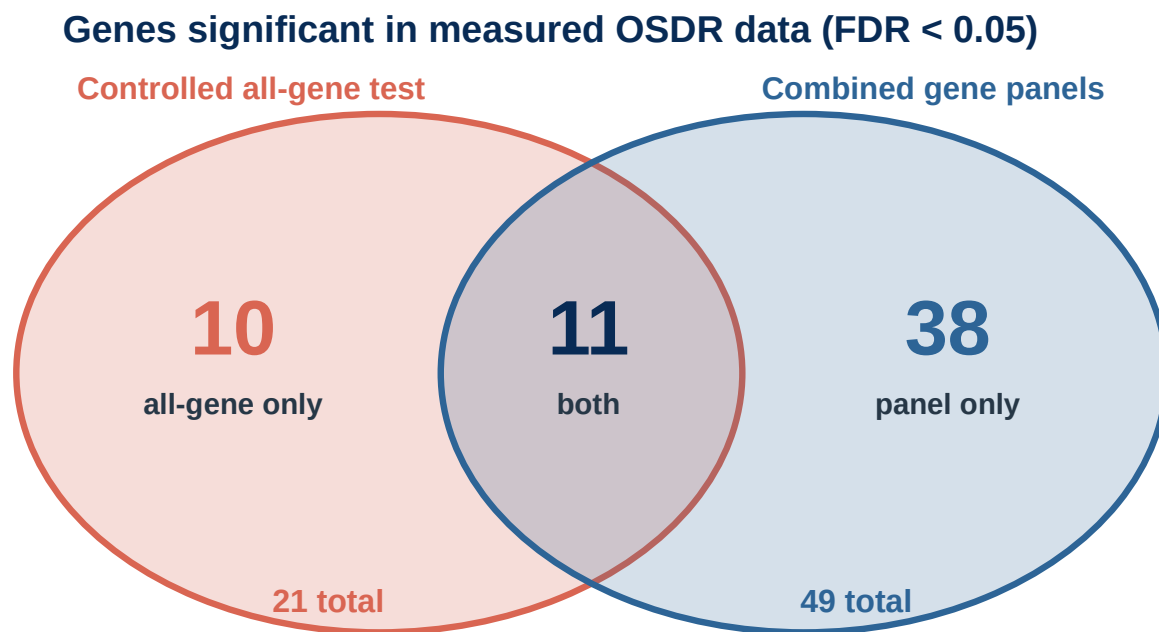


Teal improved every overall and study-average check; coral had mixed results.

FEATURE ANALYSIS

All-gene tests and compact gene panels answer different questions

Both require $FDR < 0.05$ in measured data. FDR controls false positives when many genes are tested.

**PRIMARY****Controlled all-gene test**

Only the training data change. Prediction and gene importance are tested on unseen real samples.

SECONDARY**Combined gene panel**

Combines real and synthetic rankings to retain small groups of correlated genes.

SHARED**Measured-data significance**

FLT-GC differences and false-discovery correction use measured OSDR samples only.

Thymus is strongest because both methods identify the same cell-cycle signal.

LITERATURE INTERPRETATION

How a gene was selected and what the literature says are separate

We labeled all 49 combined-panel associations by selection pattern and by prior biological evidence.

01 Fix the candidate

Gene, tissue, and FLT direction

02 Search

Spaceflight and mechanism literature

03 Compare

Observed result with published evidence

04 Record

Category, rationale, and source IDs

Selection status

26 Promoted

Selected only after adding the synthetic ranking

23 Reinforced

Selected by both real-only and combined rankings

Literature interpretation

Class	Count	Meaning
■ Aligning	22	Direct or same-tissue process agreement
■ Complementary	19	Related process or mechanism
■ Ambiguous	4	Mixed or context-dependent evidence
■ Unmatched	4	No sufficiently specific match found

Selection describes our analysis. The literature label describes previous biological evidence.

CONTROLLED-TEST COVERAGE

We ran the same controlled test in all 27 tissue analyses

Better FLT-GC prediction and interpretable genes did not always occur together.

4 Prediction + genes

All six checks improved; at least one significant gene helped prediction

TISSUES / MUSCLE GROUPS

- Liver
- Skin
- Spleen
- Thymus

14 Prediction only

All six checks improved; no gene met both criteria

TISSUES / MUSCLE GROUPS

- EDL
- Adrenal Gland
- Bone Marrow
- Brain
- Brown adipose tissue
- Cerebellum
- Colon
- Eye
- Lung
- Mammary Gland
- Optic Nerve
- Retina
- Skeletal muscle (pooled)
- White adipose tissue

9 Mixed result

At least one prediction check worsened

TISSUES / MUSCLE GROUPS

- Gastrocnemius
- Quadriceps
- Soleus
- Tibialis Anterior
- Bone
- Cecum
- Heart
- Hippocampus
- Kidney

COMBINED-RANKING GENE INVENTORY

Ten tissue analyses had genes prioritized by the combined ranking

All 49 associations were significant in real OSDR data (FDR < 0.05). Synthetic data changed priority, not significance.

Rows: FLT direction + selection status ■ aligning ■ complementary ■ ambiguous ■ unmatched

Gene color = literature | Non-empty rows shown

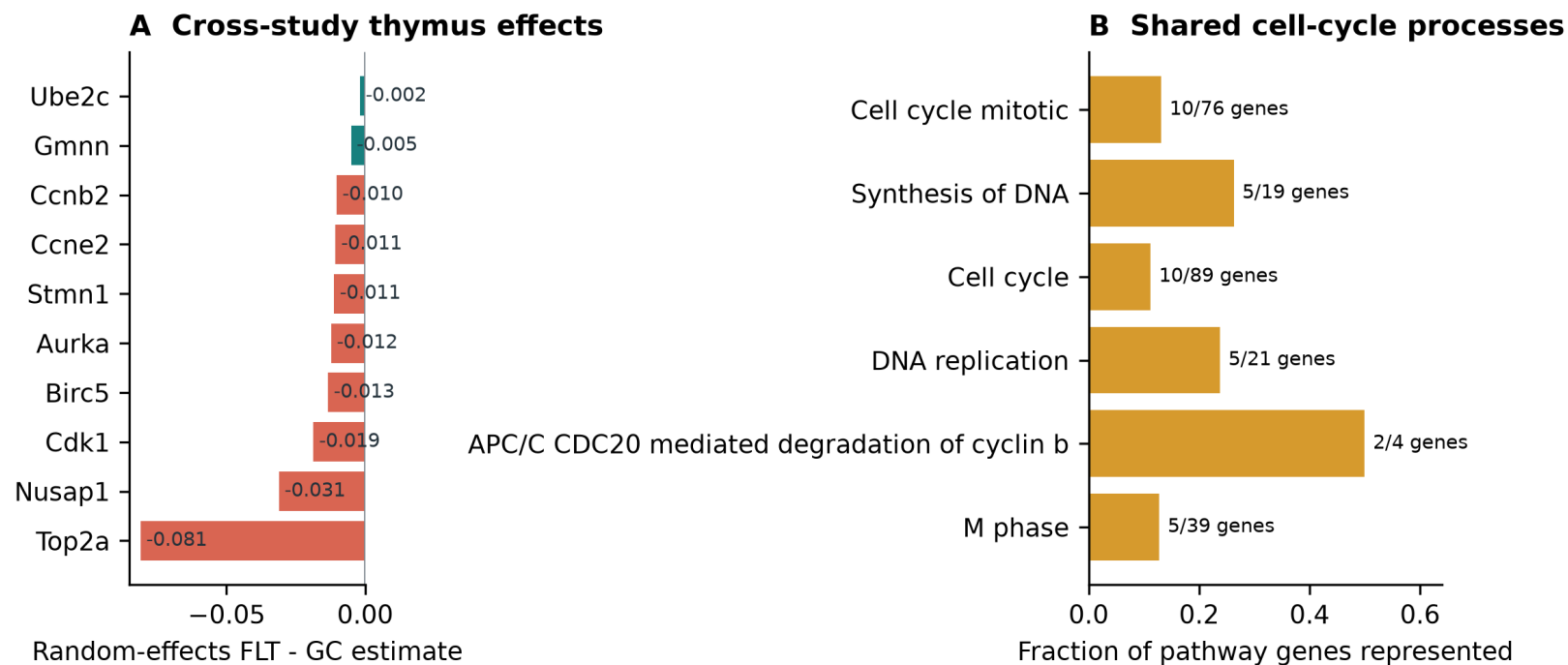
Thymus 16 genes	FLT higher	Promoted	<i>Hsd17b11</i> , <i>Etv1</i>
	FLT higher	Reinforced	<i>Snx7</i>
	FLT lower	Promoted	<i>Nusap1</i> , <i>Stmn1</i> , <i>Birc5</i> , <i>Cdk1</i> , <i>Top2a</i> , <i>Ccnb2</i> , <i>Aurka</i> , <i>Ccne2</i> , <i>Kif20a</i> , <i>Pcna</i> , <i>Ccnf</i>
	FLT lower	Reinforced	<i>Ube2c</i> , <i>Gmnn</i>
Spleen 4 genes	FLT higher	Promoted	<i>Rai14</i> , <i>Myl9</i> , <i>Ptprk</i>
	FLT higher	Reinforced	<i>Loxl1</i>
Kidney 2 genes	FLT higher	Promoted	<i>Inpp4b</i>
	FLT higher	Reinforced	<i>Slc37a4</i>
Gastrocnemius 2 genes	FLT higher	Promoted	<i>Nfkbia</i>
	FLT lower	Promoted	<i>Fhl2</i>
Eye 1 gene	FLT lower	Reinforced	<i>Klhl21</i>
Skeletal muscle (pooled) 12 genes	FLT higher	Reinforced	<i>Sox4</i> , <i>Cebpd</i> , <i>Sh3bp5</i> , <i>Prkcd</i> , <i>Arid5b</i> , <i>Sesn1</i> , <i>Tle1</i>
	FLT lower	Promoted	<i>Klhl21</i> , <i>Mapkapk5</i> , <i>Reep5</i> , <i>Itgb5</i>
	FLT lower	Reinforced	<i>Bphl</i>
Soleus 5 genes	FLT higher	Reinforced	<i>Tpm1</i>
	FLT lower	Reinforced	<i>Bdh1</i> , <i>Ech1</i> , <i>Bnip3</i> , <i>Decr1</i>
Tibialis anterior 4 genes	FLT higher	Promoted	<i>Cebpd</i>
	FLT higher	Reinforced	<i>Cdkn1a</i> , <i>St3gal5</i> , <i>Bnip3</i>
Adrenal gland 2 genes	FLT lower	Promoted	<i>Psmb8</i>
	FLT lower	Reinforced	<i>Tspan4</i>
Skin 1 gene	FLT higher	Promoted	<i>Plscr1</i>

THYMUS

Thymus is strongest across both analyses

The controlled test supports a synthetic contribution; the combined ranking adds related cell-cycle genes.

Thymus genes point to lower cell-cycle activity in flight



CONTROLLED ALL-GENE TEST

15

significant genes that
also
helped FLT-GC
prediction

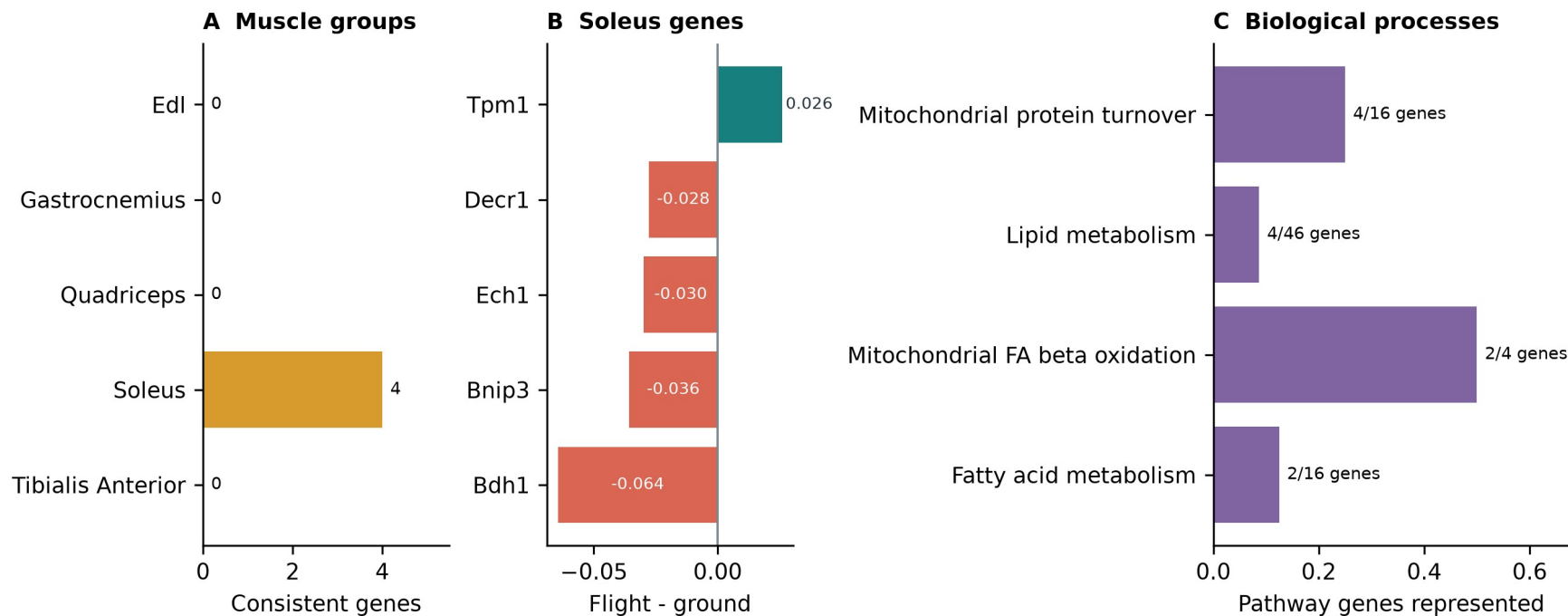
7 synthetic-added | 8 shared

- 26 significant Reactome terms; mitotic cell cycle leads
- Nine genes overlap the 16-gene combined panel
- The combined ranking adds Cdk1, Top2a, Aurka, Ccne2, Pcnf and Ccnf
- Lower mitotic renewal or fewer cycling thymocytes in flight

Soleus has a coherent secondary gene panel

The five genes form a biological story, but adding synthetic samples did not improve every prediction check.

Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis



5-gene combined panel

selected by both real and combined rankings

Prediction criteria not met

1 measure improved; 2 declined

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Panel-level hypothesis; no confirmed synthetic benefit

ADDITIONAL CONTROLLED-TEST FINDINGS

Liver, skin and spleen add narrower gene-level results

Pooled muscle improves prediction, while its gene panel remains a secondary result.

Analysis unit	Evidence source	Genes and FLT direction	Interpretation
Liver	All-gene test	Lower: Grb10, Ppic, H2-DMa, Gtf2a2	Four shared-importance candidates; no significant Reactome set enrichment.
	Combined ranking	No retained panel	
Skin	All-gene test	Higher: Plscr1	Both analyses support Plscr1; broader cell-cycle and DNA-repair context comes from the gene panel.
	Combined ranking	Higher: Plscr1 plus broader selected pathways	
Spleen	All-gene test	Higher: Loxl1	Loxl1 appears in the controlled test; the four-gene adhesion and cytoskeletal panel is tentative.
	Combined ranking	Higher: Loxl1, Rai14, Ptprk, Myl9	
Pooled muscle	All-gene test	All six checks improved; no single gene met both criteria	Adding synthetic samples improves prediction; individual genes come from the secondary ranking analysis.
	Combined ranking	12-gene interferon and sialic-acid panel	

ADDITIONAL COMBINED-RANKING FINDINGS

Several smaller panels did not pass the controlled all-gene criteria

These remain tissue-specific hypotheses from the secondary ranking analysis.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Kidney	Promoted	Inpp4b higher	Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.
	Reinforced	Slc37a4 higher	
Adrenal gland	Promoted	Psmb8 lower	No close prior match was found for these immune, proteostasis, or tissue-composition candidates.
	Reinforced	Tspan4 lower	
Gastrocnemius	Promoted	Nfkb1a higher; Fhl2 lower	NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.
	Reinforced	None	
Tibialis anterior	Promoted	Cebpd higher	Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.
	Reinforced	Cdkn1a, St3gal5, Bnip3 higher	

TAKEAWAYS

Use controlled tests to measure benefit; combined rankings to interpret biology

Synthetic profiles can help prediction and gene prioritization, but biological significance still comes from measured OSDR samples.

01 Generate

Conditional diffusion produced realistic expression profiles that were difficult to distinguish from measured samples.

02 Test

Adding synthetic profiles improved all six checks in 18 of 27 tissue analyses and retained 21 significant gene associations.

03 Interpret

The combined ranking recovered broader groups of correlated genes. Thymus was strongest across both analyses; soleus remained secondary.

Thank you

Questions?

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Mentor

James Casaletto

Program

NASA Space Life Sciences Training Program

Data

NASA OSD R | ARCHS4 | Reactome

Compute

NASA Ames Research Center

Code and
manuscript

github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSD R profiles.